nvfp4_gemm

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▼ Description

You will implement a block scaled matrix-matrix multiplication kernel optimized for NVIDIA B200. To be explicit, you will be given a tuple of tensors:

```
(a, b, sfa, sfb, c)
```

where:

- `a` is M x K x L in K-major order in nvfp4(e2m1)
- `b` is N x K x L in K-major order in nvfp4(e2m1)
- `sfa` is M x (K // 16) x L in K-major order in fp8(e4m3fnuz)
- `sfb` is N x (K // 16) x L in K-major order in fp8(e4m3fnuz)
- `c` is M x N x L in fp16

Matrix sizes `M` is divisible by mma_tiler_mn[0], `N` is divisible by mma_tiler_mn[1], `K` is divisible by 256. The ranking criteria is the geometric mean of the benchmark results. For the grand price, your kernel will be evaluated against the speed of light analysis and the solution closest to the speed of light will be awarded the grand price.

```
The speed of light analysis based on the max(FP4 Tensor Core math throughput, D
M   N    K   L time[us]
128 7168 16384 1 8.994
128 4096 7168  1 2.354
128 7168 2048  1 1.333
```

▼ Benchmark Shapes

- `{"k":16384,"l":1,"m":128,"n":7168}`
- `{"k":7168,"l":1,"m":128,"n":4096}`
- `{"k":2048,"l":1,"m":128,"n":7168}`

RANKINGS REFERENCE SUBMISSION

NVIDIA

Hide

gau.nernst	9.982µs		submission_mix.py
s.am._	10.060µs	+0.078µs	sub_v2.py
billcarson	10.137µs	+0.077µs	submission.py
shiyegao	10.616µs	+0.479µs	result.py
shigao	10.627µs	+0.011µs	node_41_plus.py
guaguabear	10.739µs	+0.112µs	final.py
rex_cz	10.748µs	+0.009µs	nvfp4_gemm_v3.py
Simon	10.807µs	+0.059µs	submission.py
yue	10.914µs	+0.107µs	submit_v1_try0.py
currybab	10.931µs	+0.017µs	submission_8.py
qinking	10.946µs	+0.015µs	p2_v1.py
XHN	10.970µs	+0.024µs	ref06.py
houdinied	10.977µs	+0.007µs	v7sub.py
shellsmile15795	11.012µs	+0.035µs	submission_v2.py
Arseni Ivanov	11.029µs	+0.017µs	blackwell_cute_2.py
__seal	11.135µs	+0.106µs	submission.py
rcmalli	11.148µs	+0.013µs	submission_static.py
snowclipsed	11.243µs	+0.095µs	nvfp4_gemm_1x4x1.py
mdouglas	11.885µs	+0.642µs	gemm3.py
rt11	11.990µs	+0.105µs	v_c.py
swanbomb_	12.135µs	+0.145µs	tcgen.py
netvope	12.223µs	+0.088µs	submission.py
vyom	12.804µs	+0.581µs	17.py
jason	12.852µs	+0.048µs	submission8.py
wolfeheart	13.221µs	+0.369µs	mission_v635_id_sweep
cdtmc	13.304µs	+0.083µs	gemmmmmmmm.py

ifndef_rust_define_ru	13.306µs	+0.003µs	roblem_first_submitted.p
lucifer_0000007	13.346µs	+0.040µs	submission.py
dxyz	13.379µs	+0.033µs	submission_v2.py
Her_77	13.398µs	+0.019µs	submission.py
Xiaozhu Meng	13.399µs	+0.001µs	's_ab5stages_c5stages.p
dandanaka_hitman	13.400µs	+0.002µs	submission_3.py
	13.408µs	+0.008µs	submission.py
Batuhanaktas	13.413µs	+0.004µs	kai_3.py
macto	13.415µs	+0.002µs	sub_ref.py
DankFarrik	13.421µs	+0.006µs	cublas.py
amackenzie-jumptrac	13.423µs	+0.001µs	submission_3.py
Carlos Andrés Chimae	13.441µs	+0.019µs	ntry.py
CatsRCool	13.837µs	+0.396µs	cuda.py
jihun_lee__	13.953µs	+0.116µs	submission.py
whoknows	14.414µs	+0.460µs	sol_prob2.py
lambda	14.597µs	+0.183µs	nvidia_gemm_soo.py
rootacess	14.738µs	+0.141µs	ubmission_cutlass_vllm.p
taka09203	16.585µs	+1.848µs	gemm_t2.py
ago.lajko	17.003µs	+0.418µs	triton_sub.py
tipsytoad__20249	17.025µs	+0.021µs	gemm_triton.py
gilsaia	17.282µs	+0.257µs	triton_5.py
shiyeegao	17.371µs	+0.088µs	node_159.py
Founzo	17.618µs	+0.247µs	submission_v1.py
pongtsu	17.679µs	+0.061µs	v4.py
mehdi	17.847µs	+0.168µs	triton_custom3.py
sahanp	19.947µs	+2.101µs	gemm.py
J	22.399µs	+2.452µs	submission.py
phuc9702	23.629µs	+1.230µs	nvfp4_gemmm.py
francescagreco	28.489µs	+4.860µs	avecKQ.py
nrehiew	28.901µs	+0.412µs	submission_naive.py

brian4983	29.689µs	+0.788µs	submission.py
Light	32.122µs	+2.433µs	cute_gemm_v0.py
Nymph	34.903µs	+2.782µs	submission.py
guojun	35.196µs	+0.293µs	submission_v1.py
ra_XOr	36.028µs	+0.832µs	GEMM_ver0.06.py
revess	36.594µs	+0.566µs	submission.py
blink1205	36.616µs	+0.022µs	submission_variant_43.p
shikhar	37.498µs	+0.882µs	nvfp4_gemm_k_tuned.p
IncompetentGeomete	37.966µs	+0.468µs	submission_cutlass.py
gup	38.954µs	+0.988µs	submission.py
Sadek Hossain Asif	44.861µs	+5.907µs	ref.py
poornaravuri	44.867µs	+0.006µs	submission.py
abao_ai	44.889µs	+0.022µs	reference.py
yasmine3457	44.896µs	+0.007µs	sub.py
whatdhack_	44.919µs	+0.023µs	submission_torch.py
Berry258	44.921µs	+0.001µs	baseline.py
andrewpeterson	44.935µs	+0.014µs	reference_submission.py
NewFreezer	44.939µs	+0.004µs	based.py
Jasmin Bogatinovski	44.949µs	+0.009µs	solution1.py
cal.culus	44.981µs	+0.032µs	nvfp4_gemm.py
olibartfast	44.982µs	+0.001µs	_gemm_submission_1700
.ryanrong	50.469µs	+5.487µs	template.py
peisong	50.515µs	+0.046µs	sol-1.py
rl5409_44612	55.200µs	+4.684µs	submission.py
DizzleRama	55.442µs	+0.242µs	erence_solution_tester_5
theeprime	56.019µs	+0.577µs	some_sol.py
mysfi	56.168µs	+0.149µs	nvfp4_gemm_class.py
ayghri	56.184µs	+0.016µs	submission.py
bargav	56.284µs	+0.100µs	template.py
sbstndbs	56.659µs	+0.375µs	submission.py

ozamatash	56.783µs	+0.123µs	nvfp4_gemm.py
siro	56.796µs	+0.014µs	submission.py
georges314	56.811µs	+0.015µs	submission.py
MonTy	56.816µs	+0.005µs	submission.py
arsrivish26691	56.822µs	+0.006µs	nvfp_cute.py
hane hane	56.828µs	+0.005µs	submission.py
Hamadullah Bijarani	56.834µs	+0.007µs	submission.py
chillparachute	56.837µs	+0.003µs	submission.py
rbyrots	56.838µs	+0.001µs	submission.py
drillyb	56.851µs	+0.014µs	submission.py
Yeargod	56.857µs	+0.006µs	submission.py
steve	56.859µs	+0.002µs	submission.py
Ali-Nawed	56.863µs	+0.003µs	submission.py
Yadui	56.879µs	+0.016µs	submission.py
jamesv.0470	56.888µs	+0.009µs	submission.py
bellz199	56.902µs	+0.014µs	submission.py
lucifer_is_back_	56.910µs	+0.008µs	gemm.py
Raghul	57.173µs	+0.262µs	submission.py
_spatters	57.501µs	+0.329µs	submission.py
stargaryen622	57.567µs	+0.066µs	submission.py
John	58.700µs	+1.133µs	submission_v2.py
Clem_Hardy	61.485µs	+2.785µs	reference.py
_hui_xu	62.821µs	+1.336µs	nvfp4_gemm.py
tarun_sempai	62.883µs	+0.062µs	nvfp4_mmul.py
muxfd	84.487µs	+21.604µs	submission.py
Hoyoun Jung	86.317µs	+1.830µs	submission.py
chthollyboss	86.395µs	+0.077µs	submission.py
cooldragon12	86.499µs	+0.104µs	submission.py
pengwh	86.765µs	+0.266µs	submission.py
juiceyjay33	87.268µs	+0.503µs	submission.py

saatwik27	87.274µs	+0.006µs	baseline.py
TLDR-Lead	581.560µs	+494.286µs	full_submission.py
Kazim	4588.002µs	+4006.442µs	nvfp4_gemm.py

Performance Trend

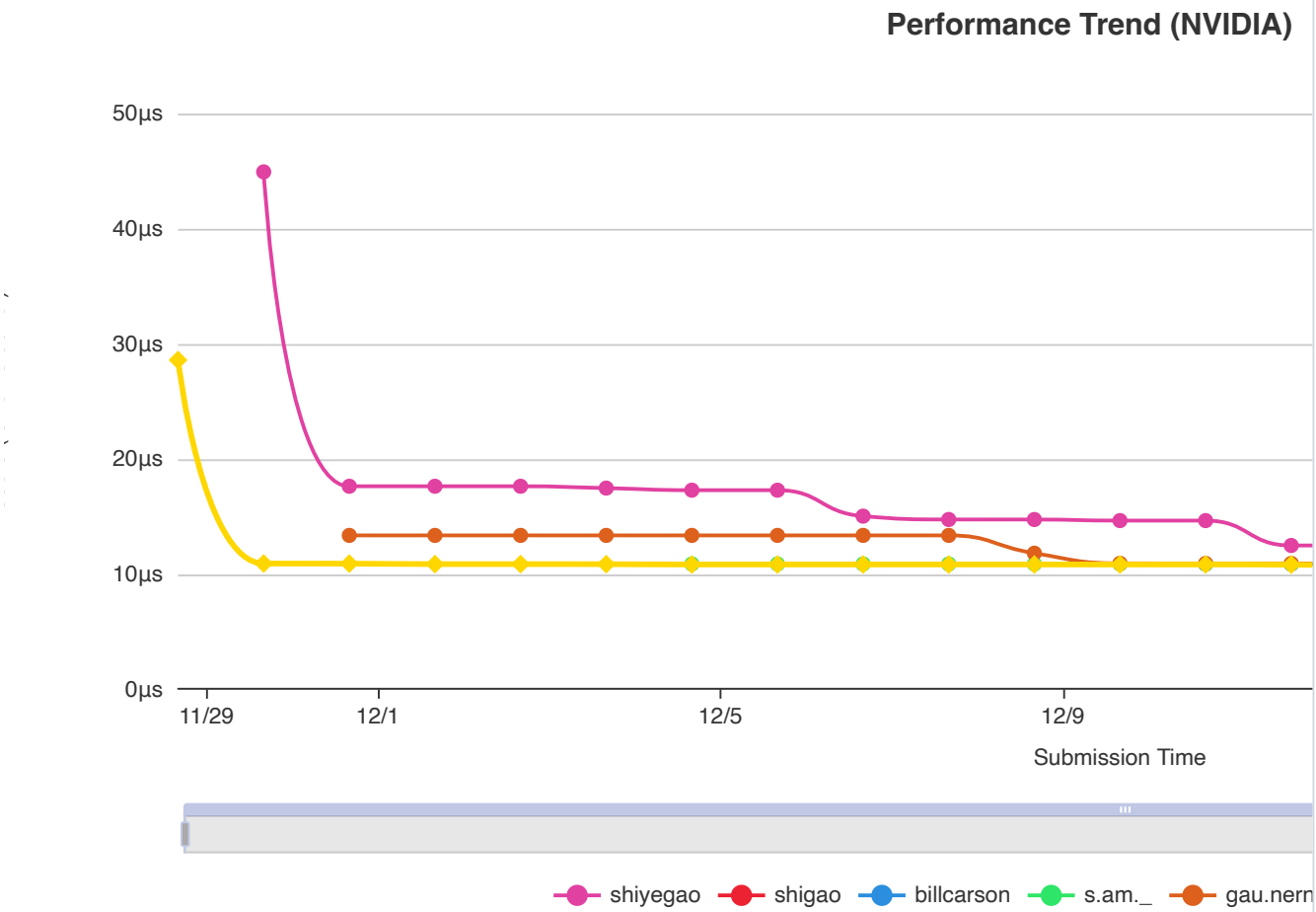
Contestant Search: gau.nernst, s.am._, billcarson, shiyegao, shigao, Type to sea...

GPU Type: NVIDIA

Load Top 5

Best Over Time

Raw Submissions



X-Axis (Date): Start mm/dd/yyyy End mm/dd/yyyy

Y-Axis (µs): Min Max

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